UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

2226720464

PHYSICS **5054/04**

Paper 4 Alternative to Practical **October/November 2008**

1 hour

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper-clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **9** printed pages and **3** blank pages.

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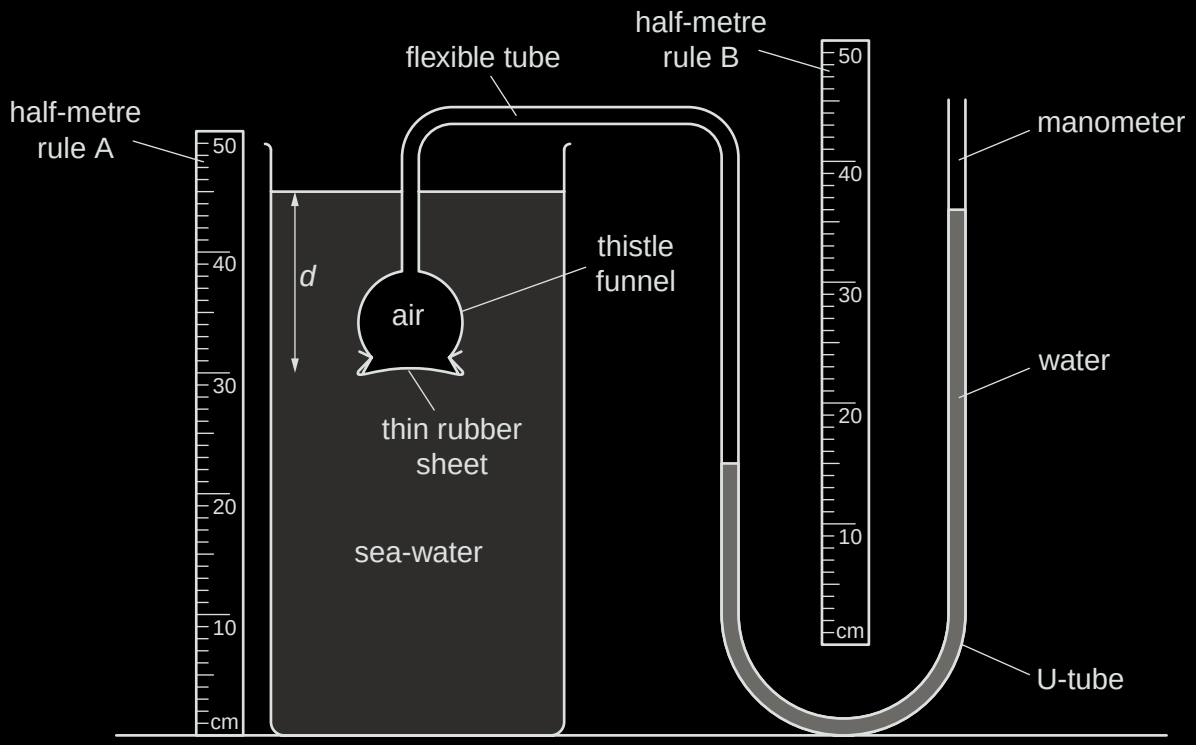


Fig. 1.1

The pressure of the air in the thistle funnel is measured with the water manometer.

The thistle funnel is lowered in the water.

(a) On Fig. 1.1, use rule B to measure the pressure shown on the manometer.

p = …… cm of water [1]

(b) Readings for p at different values of d are taken and recorded in the table of Fig.1.2.

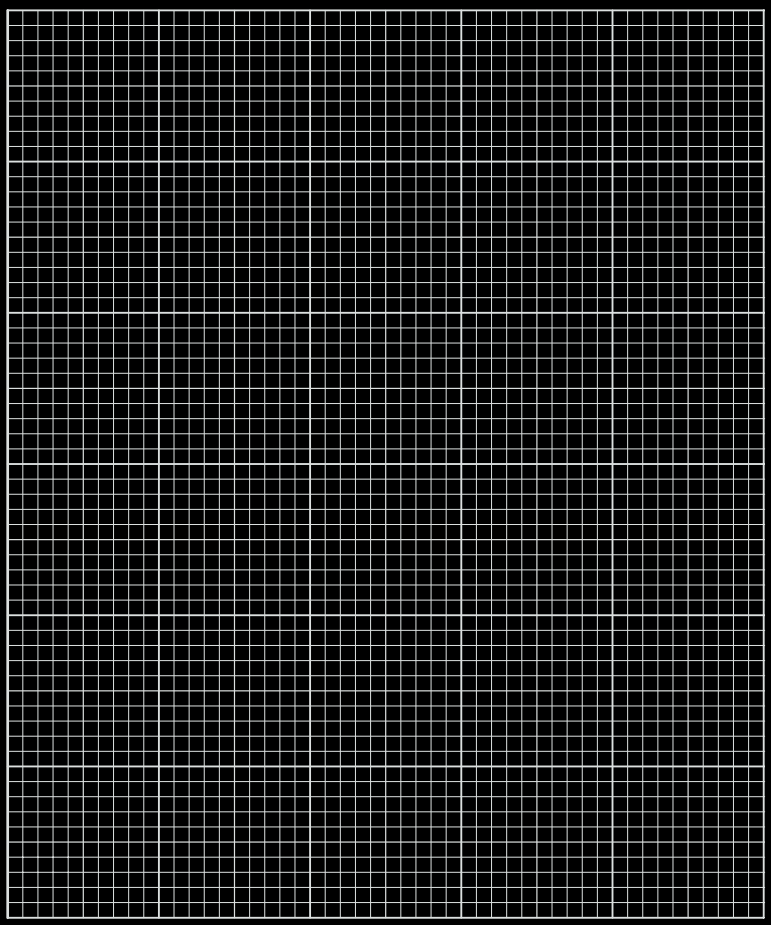
d / cm	p / cm of water
10	14
20	24
25	31
30	34
35	39
40	45

Fig. 1.2

Explain how parallax errors may be reduced when taking the readings.

.....

..... [1]



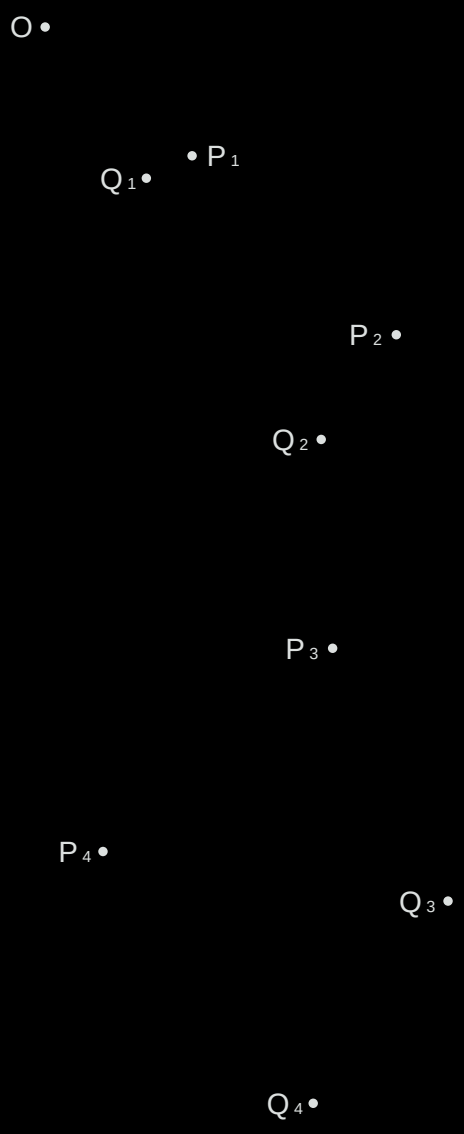


Fig. 2.1

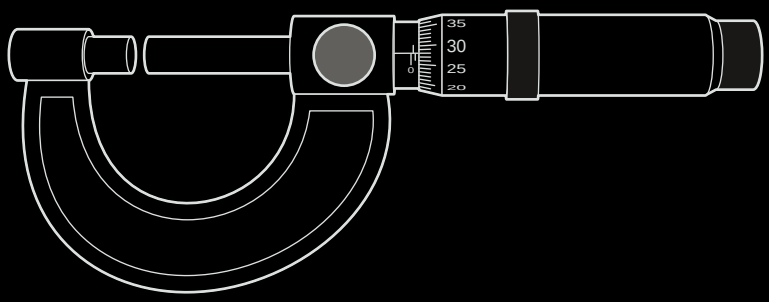


Fig. 3.1

(a) A student checks that there is no zero error on this micrometer.

Explain what is meant by a *zero error* on a micrometer.

.....

..... [1]

(b) The student uses the micrometer to determine the thickness *t* of a sheet of paper. The student folds the paper in half three times and measures the total thickness of the folded paper. The micrometer is shown in Fig. 3.2.

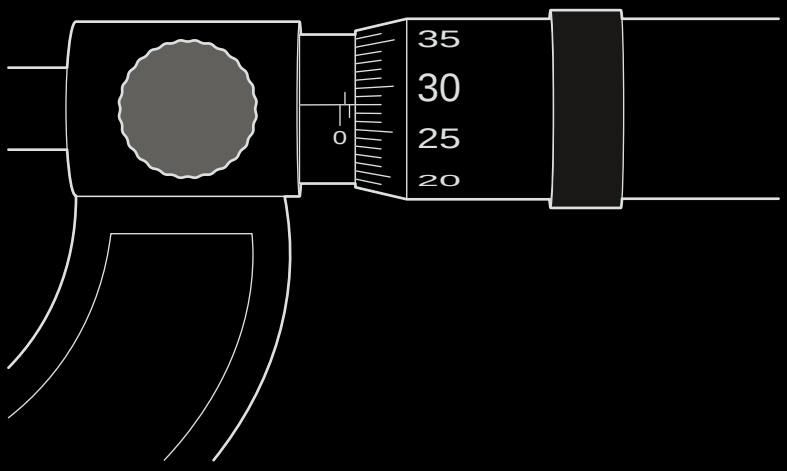


Fig. 3.2

Use Fig. 3.2 to find *t*. Show your working.

t = ... mm [3]

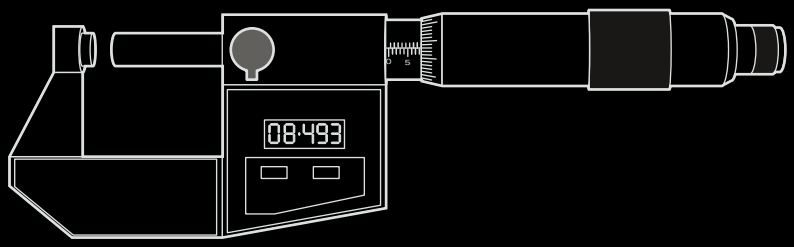


Fig. 3.3

(i) Suggest one advantage of a digital micrometer.

..... [1]

(ii) Suggest one disadvantage of a digital micrometer.

..... [1]

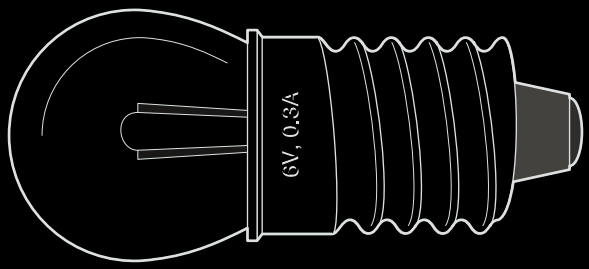


Fig. 4.1

(i) The lamp is connected in circuit A and then in circuit B, as shown in Fig. 4.2.

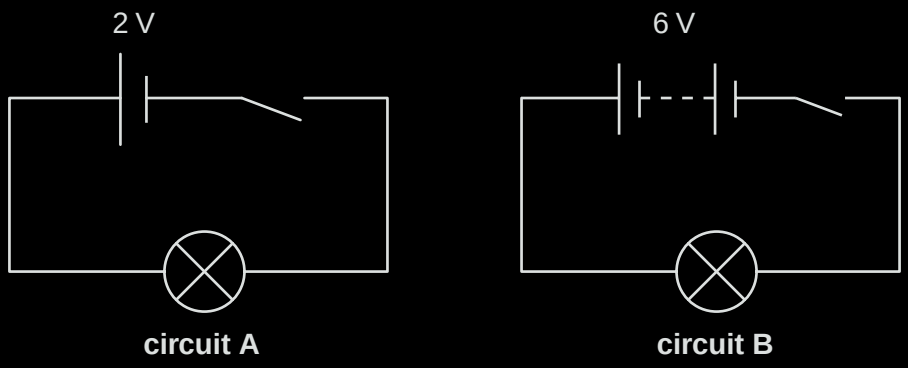


Fig. 4.2

State what the student observes when the switch is closed in each circuit.

circuit A .....

.....

circuit B .....

.....

[2]

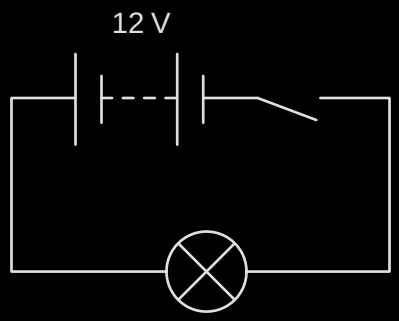


Fig. 4.3

State and explain what is observed as the switch is closed in Fig. 4.3.

.....

.....

..... [2]

(b) A lamp has no voltage marking.

Describe an experiment to find the voltage at which the lamp blows. Include a circuit diagram in your answer.

.....

.....

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.....

.....

..... [3]